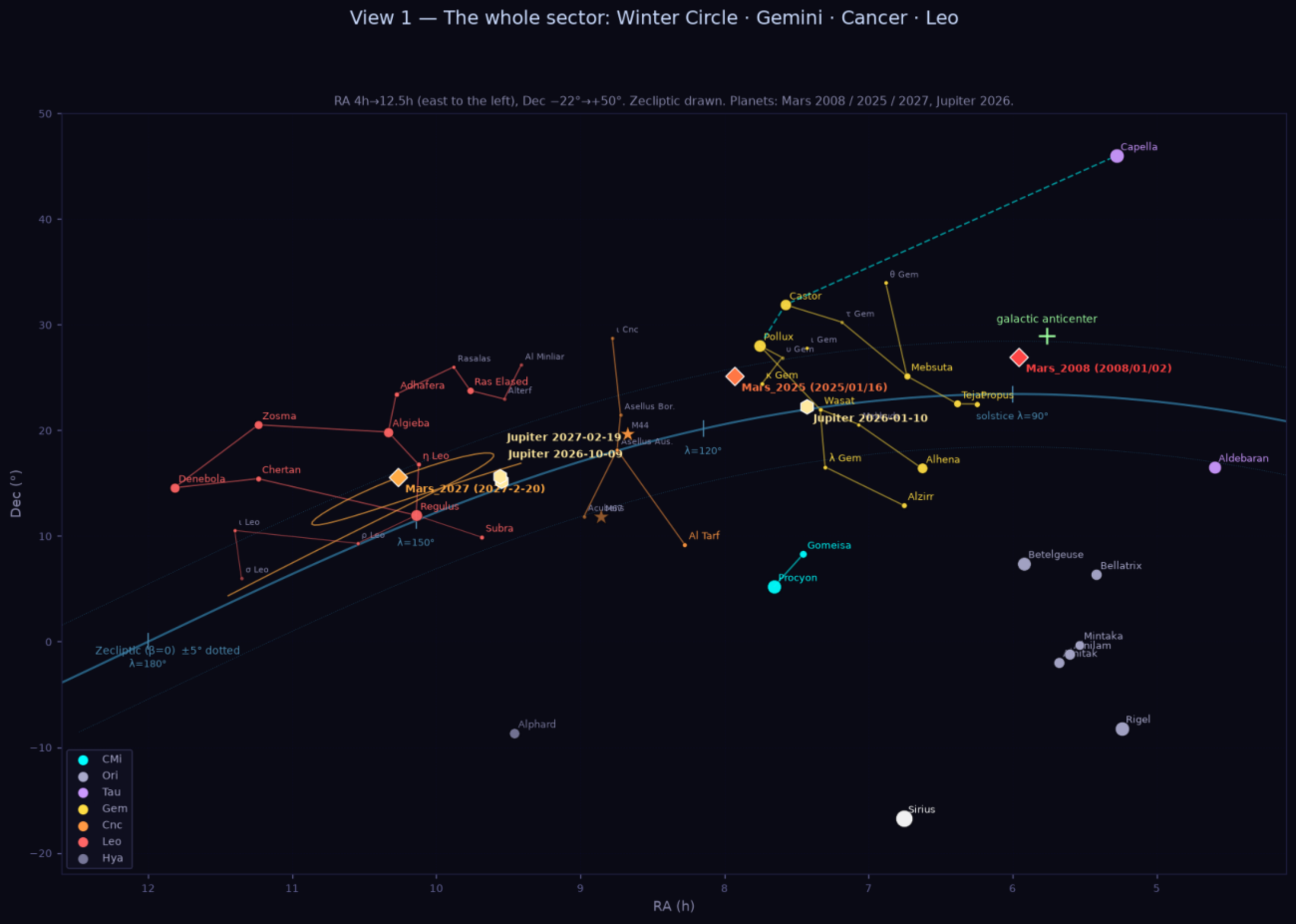
Winter Circle → Gemini → Cancer → Leo

The extension: other stars, the floor, the Lion, and Mars 2027

Dylan McCapes — October 9, 2026

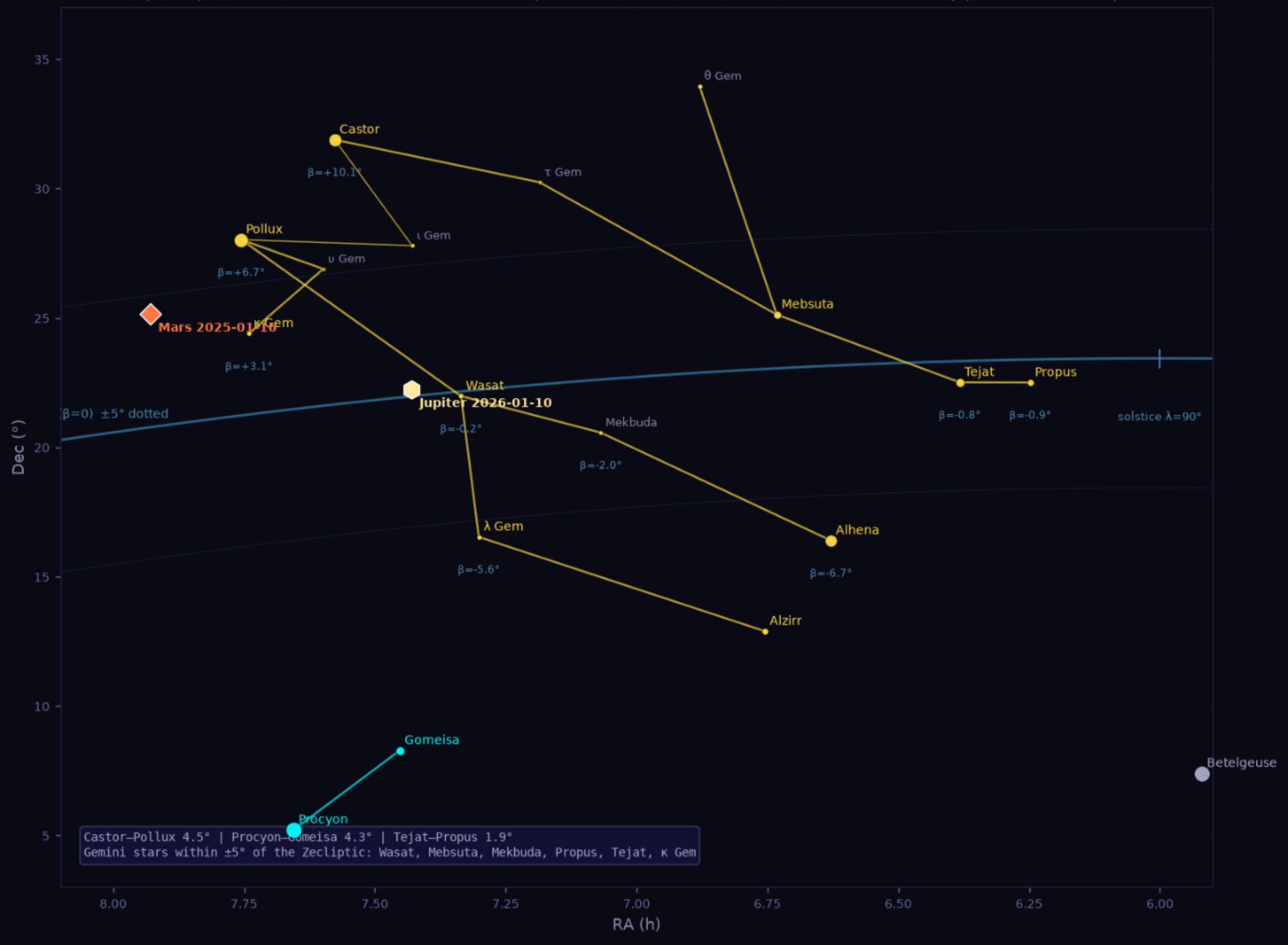
GEMINI · LEO EXTENSION — 49 stars/clusters + 3 Mars events + 3 Jupiter marks

1 The whole sector, RA 4h–12.5h, with the Zecliptic drawn
2 Gemini in full — two bodies standing on the floor
3 The dance continued from M44: what reaches Leo and what does not
4 The Zecliptic strip — λ vs β for every body
5 star_pipe over the extended set, with its structural limits stated
6 Mars 2027 — the Leo landing, computed
7 Statistics and ledger

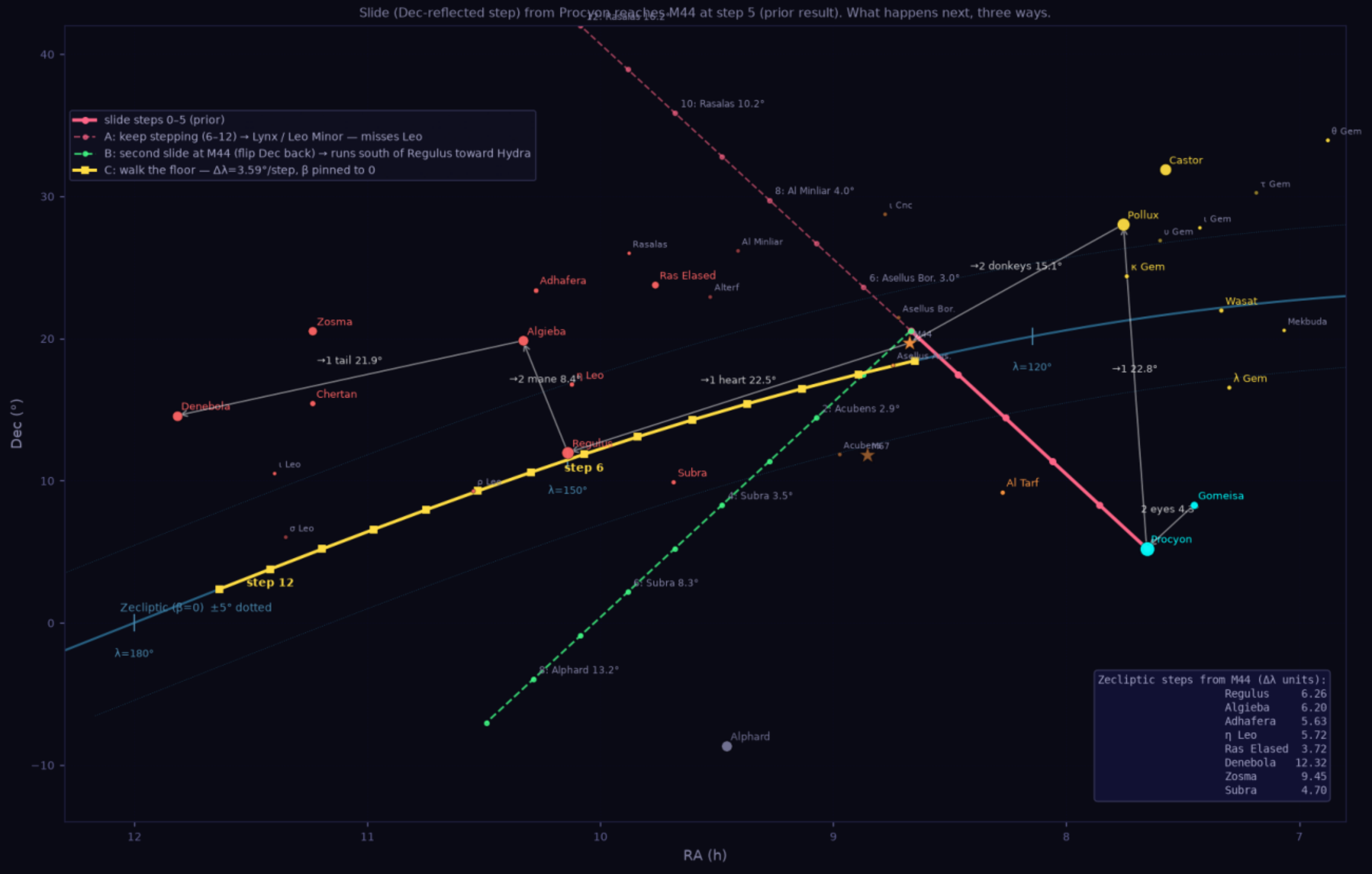

View 1 — The whole sector: Winter Circle · Gemini · Cancer · Leo
RA 4h→12.5h (east to the left), Dec −22°→+50°. Zecliptic drawn. Planets: Mars 2008 / 2025 / 2027, Jupiter 2026.
Capella
Castor
Pollux
θ Gem
τ Gem
ι Gem
υ Gem
κ Gem
Mebsuta
galactic anticenter
Mars_2008 (2008/01/02)
Mars_2025 (2025/01/16)
Wasat
TejaPropus
solstice λ=90°
Jupiter 2026-01-10
ι Cnc
Asellus Bor.
M44
Asellus Aus.
Jupiter 2027-02-19
Jupiter 2026-10-09
λ=120°
λ Gem
Alhena
Aldebaran
Alzirr
Rasalas
Al Minliar
Adhafera
Ras Elased
Alterf
Zosma
Algieba
η Leo
Denebola
Chertan
Mars_2027 (2027-2-20)
Regulus
Subra
ι Leo
ρ Leo
σ Leo
λ=150°
Acubens
Al Tarf
Gomeisa
Procyon
Betelgeuse
Bellatrix
Mintaka
Alnilam
Alnitak
Alphard
Rigel
Sirius
Zecliptic (β=0) ±5° dotted
λ=180°
CMi
Ori
Tau
Gem
Cnc
Leo
Hya
Dec (°)
RA (h)
50
40
30
20
10
0
−10
−20
12
11
10
9
8
7
6
5

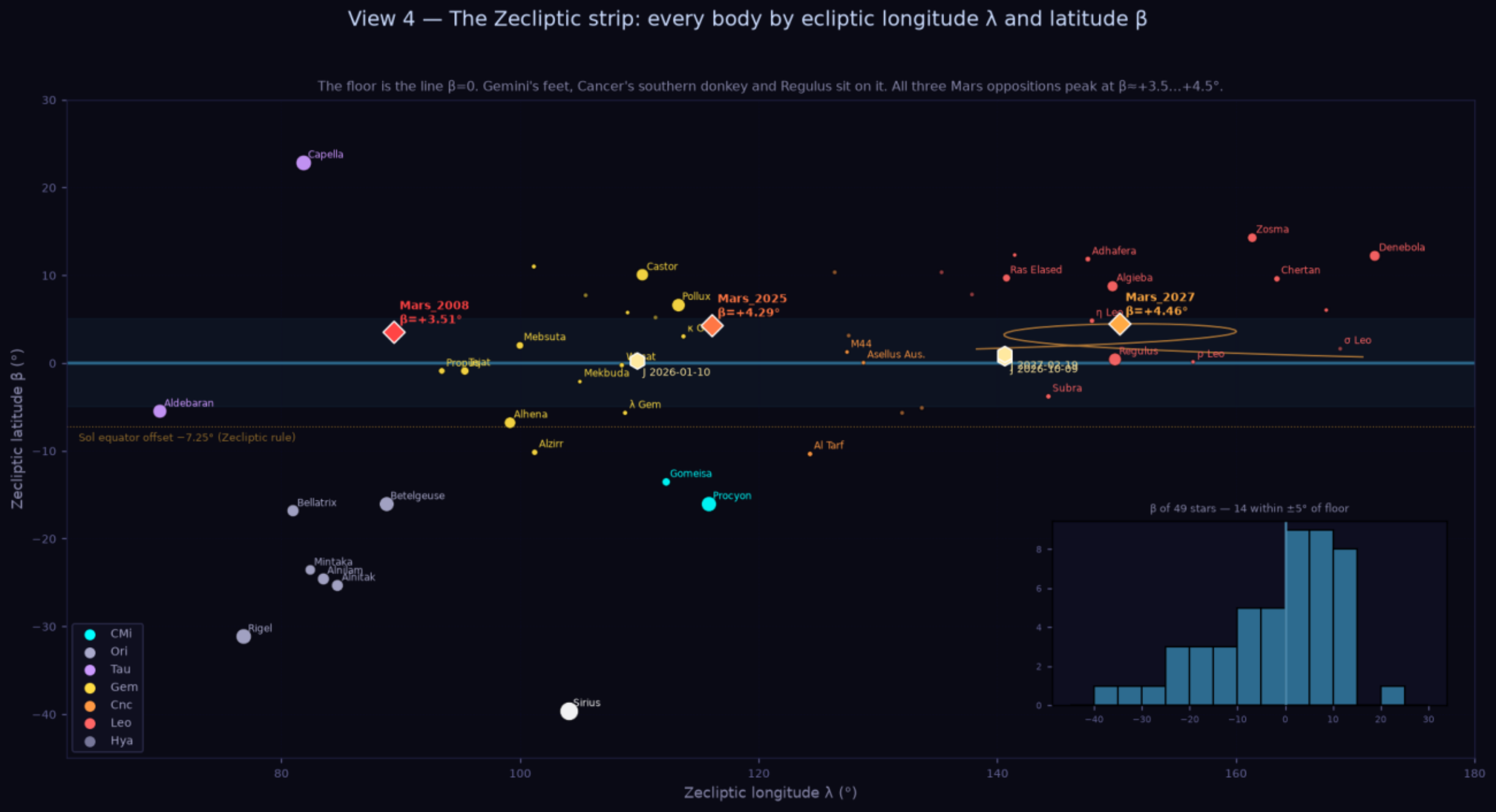
View 2 — Gemini in full: two bodies standing on the Zecliptic

Feet (Tejat, Propus, Alhena) straddle the floor near the solstice point λ=90°; heads (Castor, Pollux) are the cat's haunches. Jupiter sat in the chest Jan 2026.



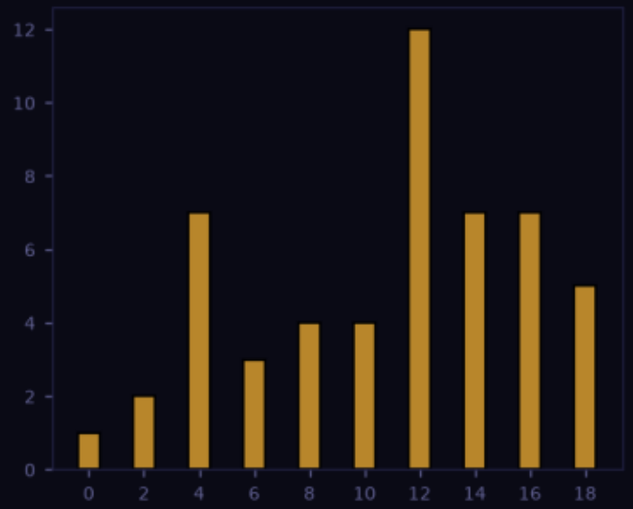
View 3 — The dance continued: from the hive into the Lion



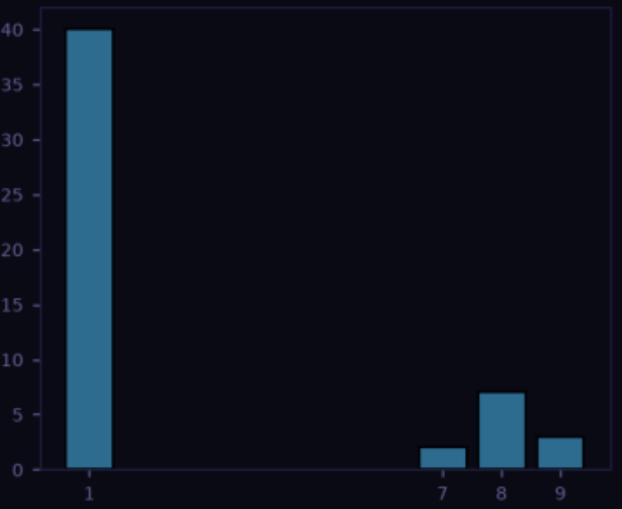

View 4 — The Zecliptic strip: every body by ecliptic longitude λ and latitude β
The floor is the line β=0. Gemini's feet, Cancer's southern donkey and Regulus sit on it. All three Mars oppositions peak at β=+3.5…+4.5°.
Zecliptic latitude β (°)
Zecliptic longitude λ (°)
Capella
Castor
Pollux
Mars_2008
β=+3.51°
Mars_2025
β=+4.29°
Mars_2027
β=+4.46°
Mebsuta
Propus
Tejat
Mekbuda
J 2026-01-10
M44
Asellus Aus.
Aldebaran
Alhena
λ Gem
Sol equator offset −7.25° (Zecliptic rule)
Alzirr
Al Tarf
Gomeisa
Procyon
Bellatrix
Betelgeuse
Mintaka
Alnilam
Alnitak
Rigel
Sirius
Zosma
Adhafera
Ras Elased
Algieba
Chertan
Denebola
η Leo
Regulus
ρ Leo
σ Leo
Subra
β of 49 stars — 14 within ±5° of floor
CMi
Ori
Tau
Gem
Cnc
Leo
Hya

View 5 — Star pipe over the extended set (same definition as WINTER_CIRCLE_PIPE_REPORT §II)

F-value distribution (F = 2·max(L,B), always even)



G (leading digit of round λ) — degenerate: every λ in 100–199 → G=1



Self-pipes (L = B): 7 of 52

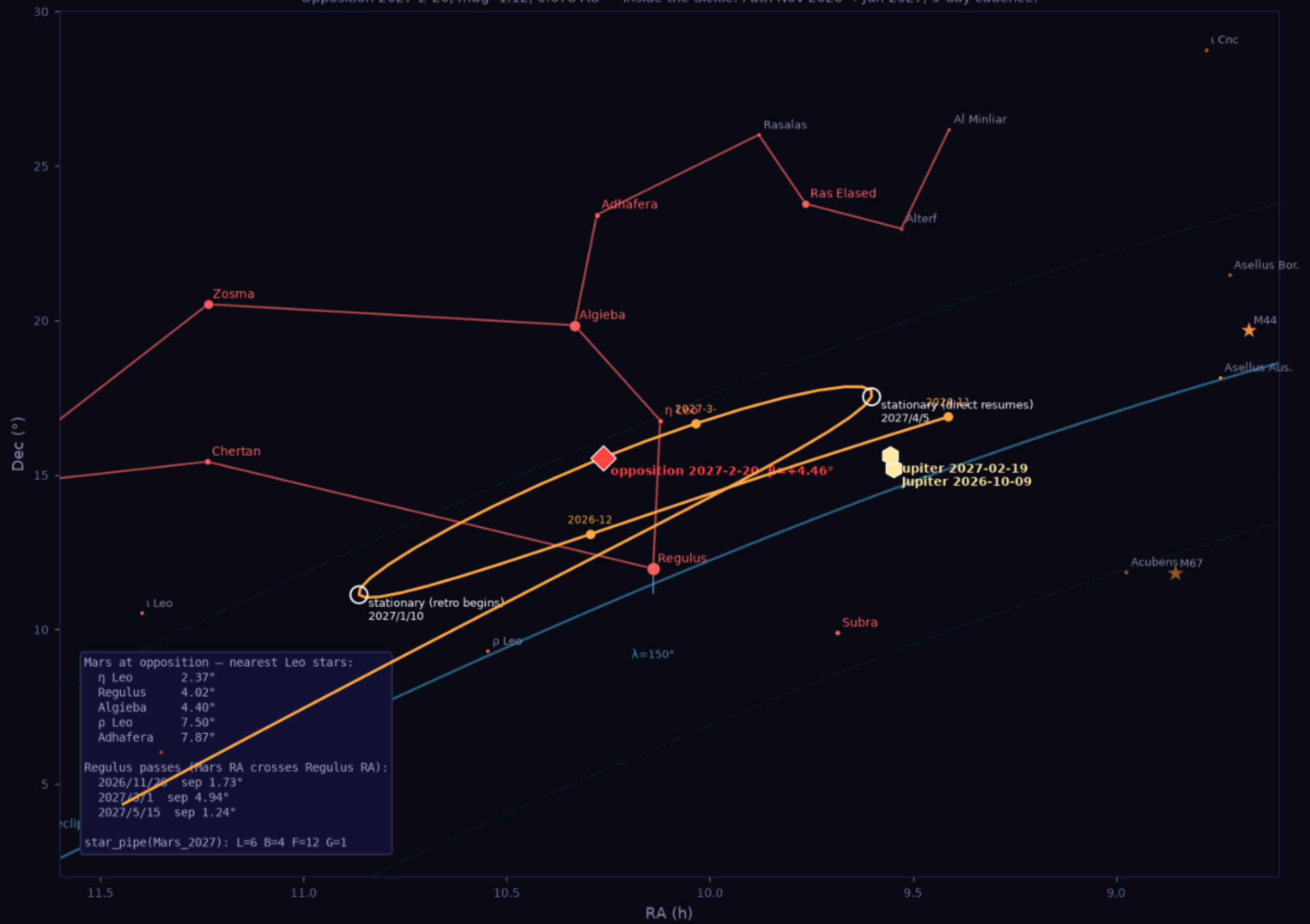
Rigel	λ= 76.8	β= -31.1	L=B=4	F=8
Gomeisa	λ= 112.2	β= -13.5	L=B=4	F=8
Betelgeuse	λ= 88.8	β= -16.0	L=B=7	F=14
Wasat	λ= 108.5	β= -0.2	L=B=0	F=0
θ Gem	λ= 101.1	β= +11.0	L=B=2	F=4
M67	λ= 132.0	β= -5.6	L=B=6	F=12
Adhafera	λ= 147.6	β= +11.9	L=B=3	F=6

F-group membership by constellation. A shared F means shared max(L,B) — the bonds are digit-sum coincidences on rounded λ/β, read as such.

	Tau	Aur	Ori	CMa	CMi	Gem	Cnc	Leo	Hya	Mars
F=0						Wasat				
F=2							M44	Chertan		
F=4						Castor Mebsuta Propus Alzirr θ Gem	Asellus Aus.	ρ Leo		
F=6							Asellus Bor.	Denebola Adhafera		
F=8			Rigel		Gomeisa	Tejat			Alphard	
F=10		Capella		Sirius		υ Gem		η Leo		
F=12			Mintaka			Mekbuda κ Gem λ Gem ι Gem	M67	Regulus Ras Elased Rasalas ι Leo σ Leo		Mars_2027
F=14	Aldebaran		Betelgeuse Alnilam Alnitak			Pollux	Al Tarf			Mars_2008
F=16					Procyon	Alhena τ Gem	Acubens	Zosma Alterf		Mars_2025
F=18			Bellatrix				ι Cnc	Algieba Subra Al Minliar		

View 6 — Mars 2027: the Leo landing (computed, PyEphem)

Opposition 2027-2-20, mag -1.12, 0.678 AU — inside the Sickle. Path Nov 2026 → Jun 2027, 5-day cadence.



GEMINI · LEO EXTENSION — 49 stars/clusters + 3 Mars events + 3 Jupiter marks

ZECLIPTIC LONGITUDE SEQUENCE (bright stars, mag < 2.6, plus floor-sitters)
  λ= 69.8 β= -5.5   Aldebaran   Tau rear_hip
  λ= 76.8 β= -31.1 Δλ= 7.0 Rigel   Ori front_paw
  λ= 80.9 β= -16.8 Δλ= 4.1 Bellatrix   Ori shoulder
  λ= 81.9 β= +22.9 Δλ= 0.9 Capella   Aur spine_apex
  λ= 82.4 β= -23.6 Δλ= 0.5 Mintaka   Ori belt_W
  λ= 83.5 β= -24.5 Δλ= 1.1 Alnilam   Ori belt_C
  λ= 84.7 β= -25.3 Δλ= 1.2 Alnitak   Ori belt_E
  λ= 88.8 β= -16.0 Δλ= 4.1 Betelgeuse   Ori chest
  λ= 93.4 β= -0.9 Δλ= 4.7 Propus   Gem Castor_foot
  λ= 95.3 β= -0.8 Δλ= 1.9 Tejat   Gem Castor_foot
  λ= 99.1 β= -6.7 Δλ= 3.8 Alhena   Gem Pollux_foot
  λ= 99.9 β= +2.1 Δλ= 0.8 Mebsuta   Gem Castor_waist
  λ= 104.1 β= -39.6 Δλ= 4.1 Sirius   CMa prey
  λ= 105.0 β= -2.0 Δλ= 0.9 Mekbuda   Gem Pollux_knee (Cepheid)
  λ= 108.5 β= -0.2 Δλ= 3.5 Wasat   Gem Pollux_waist (on Zecliptic)
  λ= 110.2 β= +10.1 Δλ= 1.7 Castor   Gem head_W / haunch_W
  λ= 113.2 β= +6.7 Δλ= 3.0 Pollux   Gem head_E / haunch_E
  λ= 115.8 β= -16.0 Δλ= 2.6 Procyon   CMi right_eye
  λ= 127.3 β= +1.3 Δλ= 11.6 M44   Cnc hive
  λ= 128.7 β= +0.1 Δλ= 1.4 Asellus Aus. Cnc donkey_S (on Zecliptic)
  λ= 147.3 β= -22.4 Δλ= 18.6 Alphard   Hya below_floor_reference
  λ= 149.6 β= +8.8 Δλ= 2.3 Algieba   Leo mane / sickle
  λ= 149.8 β= +0.5 Δλ= 0.2 Regulus   Leo heart (ON Zecliptic)
  λ= 156.4 β= +0.1 Δλ= 6.6 ρ Leo   Leo chest
  λ= 161.3 β= +14.3 Δλ= 4.9 Zosma   Leo hip
  λ= 168.7 β= +1.7 Δλ= 7.4 σ Leo   Leo rear_paw
  λ= 171.6 β= +12.3 Δλ= 2.9 Denebola   Leo tail
  sector span: 69.8° → 171.6° = 101.8° of Zecliptic (prior report: 46.0° for the Circle alone)

ON THE FLOOR (|β| ≤ 2.5°): 10
  Asellus Aus. β=+0.08° λ=128.7° Cnc
  ρ Leo   β=+0.15° λ=156.4° Leo
  Wasat   β=-0.18° λ=108.5° Gem
  Regulus   β=+0.46° λ=149.8° Leo
  Tejat   β=-0.82° λ=95.3° Gem
  Propus   β=-0.89° λ=93.4° Gem
  M44   β=+1.29° λ=127.3° Cnc
  σ Leo   β=+1.70° λ=168.7° Leo
  Mekbuda   β=-2.04° λ=105.0° Gem
  Mebsuta   β=+2.07° λ=99.9° Gem
  Regulus is the brightest star in the sky within 0.5° of the Zecliptic (β=+0.46°). The Moon occults it; Mars passes it ~every 2 years.

MARS OPPOSITIONS IN THIS SECTOR
  Mars_2008 2008/01/02 λ= 89.40 β=+3.51 mag -1.38 nearest Propus 6.0° F=14 G=8
  Mars_2025 2025/01/16 λ=116.05 β=+4.29 mag -1.25 nearest κ Gem 2.7° F=16 G=1
  Mars_2027 2027-2-20 λ=150.24 β=+4.46 mag -1.12 nearest η Leo 2.4° F=12 G=1
  Mars 2027 stations: 2027/1/10 (RA max) → 2027/4/5 (RA min); retrograde loop inside the Sickle
  Regulus passes: 2026/11/26 (1.7°); 2027/3/1 (4.9°); 2027/5/15 (1.2°)
  β at opposition: +3.51 (2008, Gemini/chest) · +4.29 (2025, Gemini/haunch) · +4.46 (2027, Leo/Sickle). Mars orbits 1.85° off the ecliptic; near opposition its β is largest because Earth is closest

JUPITER
  Jupiter_2026-01-10 (opposition, Gemini)   RA 7.43h Dec +22.2 λ=109.8 β=+0.26 nearest Wasat 1.3°
  Jupiter_2026-10-09 (today)   RA 9.55h Dec +15.2 λ=140.6 β=+0.62 nearest Subra 5.7°
  Jupiter_2027-02-19 (Mars opposition day)   RA 9.56h Dec +15.6 λ=140.6 β=+1.06 nearest Subra 6.0°

DANCE — continuation from M44
  step vector (Gomeisa→Procyon): ΔRA=+0.2025h ΔDec=-3.064°; Zecliptic Δλ=3.59°
  A. keep the reflected step past M44:
     step 6 RA 8.87h Dec +23.6 nearest Asellus Bor. 3.0°

```
   step  7  RA 9.07h Dec +26.7  nearest ι Cnc 4.4°
   step  8  RA 9.28h Dec +29.7  nearest Al Minliar 4.0°
   step  9  RA 9.48h Dec +32.8  nearest Al Minliar 6.7°
   step 10  RA 9.68h Dec +35.9  nearest Rasalas 10.2°
   step 11  RA 9.88h Dec +38.9  nearest Rasalas 12.9°
   step 12  RA 10.09h Dec +42.0  nearest Rasalas 16.2°
B. second slide at M44 (Dec flipped back):
   step  1  RA 8.87h Dec +17.5  nearest Asellus Aus. 1.9°
   step  2  RA 9.07h Dec +14.4  nearest Acubens 2.9°
   step  3  RA 9.28h Dec +11.4  nearest Acubens 4.4°
   step  4  RA 9.48h Dec +8.3  nearest Subra 3.5°
   step  5  RA 9.68h Dec +5.2  nearest Subra 4.7°
   step  6  RA 9.88h Dec +2.2  nearest Subra 8.3°
   step  7  RA 10.09h Dec -0.9  nearest Alphard 12.1°
   step  8  RA 10.29h Dec -4.0  nearest Alphard 13.2°
   step  9  RA 10.49h Dec -7.0  nearest Alphard 15.4°
→ Neither A nor B enters Leo. The slide is a Dec-dominant move; Leo lies along λ at β≈0. The phrase changes: from stepping to walking the floor.
C. walk the floor from M44 in Δλ units (β pinned to 0):
   Regulus      6.26 steps   (β of star +0.46°)
   Algieba      6.20 steps   (β of star +8.81°)
   Adhafera     5.63 steps   (β of star +11.87°)
   η Leo        5.72 steps   (β of star +4.87°)
   Ras Elased   3.72 steps   (β of star +9.72°)
   Denebola    12.32 steps   (β of star +12.27°)
   Zosma        9.45 steps   (β of star +14.33°)
   Subra        4.70 steps   (β of star -3.76°)
D. phrase lengths 2-1-2-1-2-1:
   2 eyes          Gomeisa → Procyon      4.3°
   →1              Procyon → Pollux      22.8°
   →2 donkeys       Pollux → M44         15.1°
   →1 heart            M44 → Regulus     22.5°
   →2 mane         Regulus → Algieba      8.4°
   →1 tail         Algieba → Denebola    21.9°
twin pairs 2.5°–5.5° apart in the set: 28
   2.73°  Ras Elased – Rasalas
   2.74°  Alnitak – Mintaka
   3.15°  κ Gem – υ Gem
   3.33°  Asellus Bor. – Asellus Aus.
   3.34°  Ras Elased – Alterf
   3.59°  Alterf – Al Minliar
   3.63°  Pollux – κ Gem
   3.66°  Algieba – Adhafera
   3.95°  Alhena – Alzirr
   3.99°  Wasat – Mekbuda
   4.02°  ι Gem – τ Gem
   4.30°  Algieba – η Leo
   4.30°  Procyon – Gomeisa
   4.33°  Pollux – ι Gem
   4.51°  Castor – Pollux
   4.52°  Castor – ι Gem
   4.55°  ι Leo – σ Leo
   4.80°  Regulus – η Leo
   5.00°  Castor – υ Gem
   5.09°  Zosma – Chertan
   5.22°  Mekbuda – λ Gem
   5.28°  Castor – τ Gem
   5.37°  Ras Elased – Al Minliar
   5.38°  θ Gem – τ Gem
   5.41°  κ Gem – ι Gem
   5.44°  Chertan – ι Leo
   5.46°  Wasat – λ Gem
   5.46°  Mebsuta – Tejat
```

STRUCTURAL FACTS ABOUT star_pipe (hold for every input)
  T1 F = L + |L−B| + B = 2·max(L,B). F is always even; an F-group is 'same max digit-sum'. F=14 <=> max(L,B)=7.
      (prior report's 'self-pipe stars have even F' is true but weaker: ALL F are even.)
  T2 G = leading digit of round(λ). For λ in 100–199, G=1 regardless. Gemini/Cancer/Leo all collapse to G=1; G_diff=0 there carries no information.
  T3 L and B are digit sums of ROUNDED degrees; a star 0.5° off changes L by up to 9. The pipe is a coarse hash of position, not a geometric property.
  T4 Self-pipe <=> digit-sum(round λ) ≡ digit-sum(round |β|) (mod 10). Expected rate ≈ 1/10 under independence; observed 7/49.

WHAT IS REAL GEOMETRY (independent of any pipe; all numbers computed)
  • Propus: 3.5° from the June-solstice point (λ=90°), 9.2° from the galactic anticenter.
  • Tejat: 5.4° from the June-solstice point (λ=90°), 10.6° from the galactic anticenter.
  • Floor-sitters |β| ≤ 0.5°: Wasat (λ 108.5) → Asellus Aus. (λ 128.7) → Regulus (λ 149.8) → ρ Leo (λ 156.4); Δλ: 20.2°, 21.1°, 6.6°
  • Aldebaran (λ 69.8°) and Regulus (λ 149.8°) — two of the four 'royal stars' — bracket this sector 80.0° apart.
  • In λ, Regulus sits 22.5° past M44 and 21.8° before Denebola — the heart is the λ-midpoint of hive and tail to within 1°.
  • The three long legs of the phrase (Procyon→Pollux, M44→Regulus, Algieba→Denebola) are 22.8°, 22.5°, 21.9° — spread 0.9°. Targets were chosen by name, so treat this as a curiosity, not a test.
  • Mars oppositions recur in the same sector every ~15–17 years (2008 Gem → 2025 Gem → 2027 Leo → ~2042 Gem/Cnc est.).

LEDGER coordinates: J2000 catalogue values (Hipparcos/SIMBAD-level, ±0.01°). Planets: PyEphem VSOP87 (±1′). Ecliptic transform: exact rotation, ε=23.4393°.
        Dance geometry: linear RA/Dec stepping (as in dance_projection.py) for A/B; exact ecliptic for C. Pipe: digit arithmetic, deterministic.